ICT2142 E-Business Systems

Lecture 08

Logistics, Fulfilment & Supply Chain in E-Business

Department of ICT
Faculty of Technology
University of Ruhuna



Learning Objectives

- Explain the stages of the order fulfilment process and the main fulfilment models used in e-business.
- Describe the role of inventory management and key techniques (ABC analysis, JIT, EOQ, safety stock).
- Identify inventory and warehouse technologies such as barcode scanning, RFID and WMS platforms.
- Compare delivery options available to online retailers and their cost/speed trade-offs.
- Explain tracking technologies that give businesses and customers real-time shipment visibility.
- Describe how fulfilment, inventory and delivery combine into an end-to-end e-business supply chain.

Supply Chain in E-Business

- Logistics and supply chain management covers every process that moves a product from a supplier to a warehouse and finally to the customer's door.
- In e-business, this chain is triggered the moment a customer places an online order, and it only ends when the order (or a return) is fully resolved.
- Core components: procurement and suppliers, inventory and warehousing, order fulfilment, transportation and delivery, and reverse logistics (returns).
- Logistics performance directly affects delivery speed, cost, order accuracy, customer satisfaction and repeat purchases.

Key idea: *in e-business, the supply chain doesn't stop at the sale — it continues until the product, or a refund, safely completes its journey.*

Purchasing, Logistics & Support Activities

- **Purchasing Activities -** Include identifying vendors, evaluating vendors, selecting specific products, placing orders.
- **Supply Chain -** A network between a company and its suppliers to produce and distribute a specific product, and the supply chain represents the steps it takes to get the product or service to the customer.
- **Procurement -** Includes all purchasing activities, plus monitoring of all elements of purchase transactions.
- **Sourcing-** Procurement activity devoted to identifying suppliers and determining their qualifications.
- **E-procurement or e-sourcing -** Use of Internet technologies in procurement and sourcing activities.

Direct vs. Indirect Materials Purchasing

- **Direct materials**

  Materials that become part of the finished product in a manufacturing process.

- **Indirect materials**

  Other materials that the company purchases, including factory supplies.

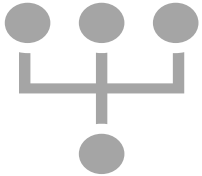
- **Replenishment purchasing**

  The company negotiates long-term contracts for most of the materials that it will need.

Supply Chain Management



Used to add value in benefits to the ultimate consumer at the end of supply chain.



Tier one suppliers

Develop long-term relationships with large number of suppliers.

Tier 1 companies, are the firms which have a direct relationship with and sell to OEMs.

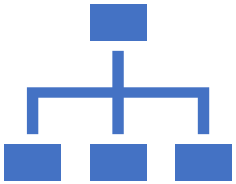
These companies consist of the most advanced technologies and processes, and are major companies in their own rights.

Supply Chain Management

- **Tier two suppliers**
  - Tier 2 refers to companies that produce and supply parts from the material obtained via Tier 3 to Tier 1 level.
  - Tier 2 firms are comparatively smaller than Tier 1 but equally vital for the supply chain.
  - They are integral to the speed of production. They must adhere to the safety and quality standards; otherwise, their products cannot go forth to Tier 1 companies.



Supply Chain Management



Tier three suppliers

Tier 3 suppliers are the foundation of the entire supply chain.

They provide the required materials, such as metals and plastic, in their raw form or almost raw state to Tier 2 and Tier 1 companies.



Supply alliances

Long-term relationships created among participants in the supply chain



Internet Technologies and the Supply Chain

Key element of successful supply chain management

Clear communications and quick responses.

Major disadvantage of using Internet technologies in supply chain management

The cost of the technologies.



Advantages of using Internet Technologies in Supply Chain Management

Suppliers can:

- Share information about customer demand fluctuations.
- Receive rapid notification of product design changes and adjustments.
- Provide specifications and drawings more efficiently.
- Increase the speed of processing transactions.
- Reduce the cost of handling transactions.
- Reduce errors in entering transaction data.
- Share information about defect rates and types.

Building and Maintaining Trust in the Supply Chain



Major issue for most companies

Developing trust.



Key elements for building trust

Continual communication and information sharing.

Logistic Activities

What Are Logistics?

- Logistics refers to the overall process of managing how resources are acquired, stored, and transported to their final destination.

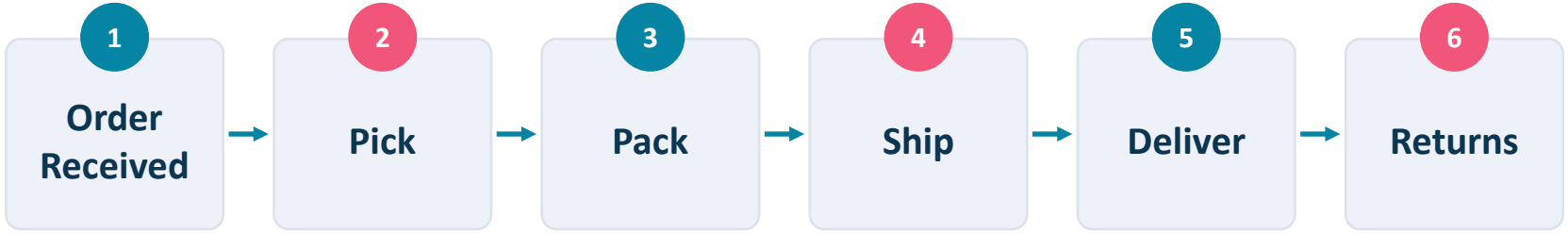
Objective of logistics

- To provide the right goods in the right quantities in the right place at the right time.

Include managing

- Inbound movements of materials and supplies.
- Outbound movements of finished goods and services.
- Logistic Activities: Receiving, warehousing, controlling inventory, scheduling and controlling vehicles, distributing finished goods.

The Order Fulfilment Process



Each stage must be fast and accurate — delays or errors at any single stage increase delivery time and reduce customer trust.

Order Fulfilment Models



In-House Fulfilment

The business stores, picks, packs and ships its own orders. Gives full control over quality and branding, but requires space, staff and investment.



Third-Party Logistics (3PL)

An external provider handles warehousing, picking and shipping. Reduces overhead and scales easily, with less direct control.



Dropshipping

The supplier ships directly to the customer; the retailer never holds stock. Low start-up cost, but thinner margins and less fulfilment control.



Fulfilment Centers

Retailer stock is stored in a shared warehouse network (e.g. marketplace-managed) for fast, standardized picking and shipping, for a fee.

Inventory Management Systems

- Inventory management is the process of tracking, controlling and optimizing stock levels across warehouses and sales channels.
- Core concepts include the SKU (stock-keeping unit), reorder point, safety stock and inventory turnover rate.
- The goal is to balance two risks: stockouts, which lose sales and damage trust, and overstocking, which ties up capital and storage space.
- An Inventory Management System (IMS) is software that centralizes real-time stock data across the online store, warehouse and suppliers.

Example: *when a customer completes checkout, the IMS instantly reduces the available stock count so the item is not oversold on other channels.*

Inventory Management Techniques



ABC Analysis

Classifies inventory by value and importance — ‘A’ items are high-value/low-quantity and need close control, ‘C’ items are low-value/high-quantity.



Just-In-Time (JIT)

Orders stock to arrive close to when it is actually needed, minimizing holding cost and warehouse space.



Economic Order Quantity (EOQ)

A formula-based approach that calculates the order size which minimizes combined ordering and holding costs.



Safety Stock & Reorder Point

A buffer quantity and trigger level that protect against stockouts caused by demand spikes or supplier delays.

Inventory & Warehouse Technologies

- **Barcode scanning:** a fast, low-cost way to identify items and update stock counts at receiving, picking and shipping.
- **RFID (Radio-Frequency Identification):** tags allow bulk, non-line-of-sight scanning for faster and more accurate stock counts.
- **Warehouse Management System (WMS):** software that directs put-away, optimizes picking routes and tracks exact stock locations.
- **Cloud-based inventory platforms:** synchronize stock levels in real time across the online store, marketplaces and physical locations.

Key idea: *connecting the IMS/WMS directly to the e-commerce platform is what actually prevents overselling out-of-stock items.*

Delivery Systems in E-Business

- Standard/economy shipping: the lowest cost option, with a longer delivery window — suited to non-urgent orders.
- Express and same-day delivery: a premium, faster service for urgent or local orders, usually at a higher price.
- Click-and-collect (BOPIS — Buy Online, Pick-up In-Store): customers order online and collect in-store or from a locker, cutting last-mile cost.
- Last-mile delivery: the final leg from the local hub to the customer's door — typically the most expensive and complex part of the journey.
- Delivery partners: businesses may use third-party couriers, postal services, or an in-house delivery fleet, each with different cost/control trade-offs.

Delivery Tracking Technologies



GPS Tracking

Gives the real-time location of delivery vehicles and parcels, enabling live route visibility and accurate ETA updates.



RFID & IoT Sensors

Monitor the location and condition of goods in transit — including temperature-sensitive shipments.



Barcode / QR Scanning

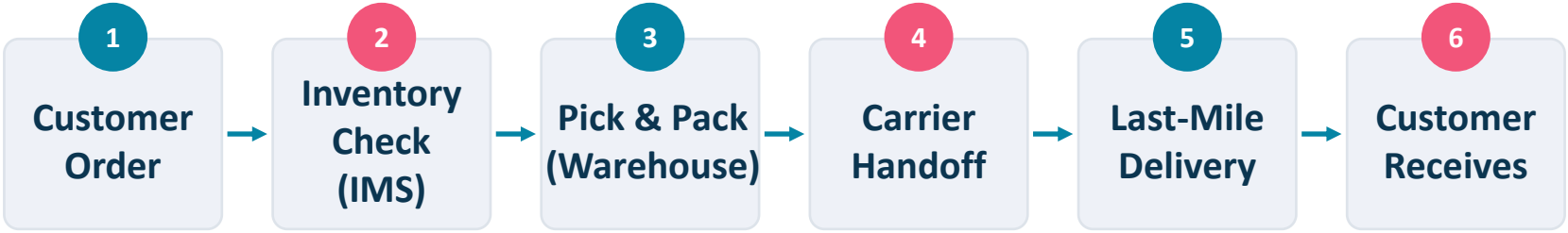
Scan events at every handoff point (warehouse, courier, depot, door) automatically update the shipment's status.



Automated Notifications

SMS, email and app alerts keep customers informed at each milestone, from dispatch to out-for-delivery to arrival.

End-to-End Logistics & Delivery Flow



Tracking data flows alongside the physical shipment at every step, giving both the business and the customer continuous visibility.

Key Challenges & Best Practices

Common Challenges

- Forecasting demand accurately for each product
- Stockouts and overstocking across channels
- Rising last-mile delivery costs
- Delivery delays and failed deliveries
- Managing returns (reverse logistics)
- Keeping stock data in sync across channels

Best Practices

- Maintain real-time inventory visibility across all channels
- Offer multiple fulfilment and delivery options
- Automate with WMS / transport management systems
- Track shipments proactively and communicate early
- Design an efficient, customer-friendly returns process

Summary & Key Takeaways

- Order fulfilment turns an online order into a delivered product through receiving, picking, packing, shipping and returns handling.
- Inventory management systems and techniques (ABC analysis, JIT, EOQ, safety stock) keep stock accurate, available and cost-efficient.
- Barcode, RFID and WMS/cloud technologies connect inventory data in real time across warehouses and sales channels.
- Delivery systems range from standard shipping to same-day and click-and-collect, each with its own cost and speed trade-offs.
- Tracking technologies — GPS, RFID/IoT, scanning and automated notifications — give visibility that builds customer trust.
- Together, fulfilment, inventory and delivery form the e-business supply chain that determines how reliably a business delivers on its promises.

Thank you… !